Ryan Tsai

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Education

University of Auckland – Bachelor of Engineering (Honours), Software Engineering Expected Nov 2027

- **GPA:** 8.25 / 9.00
- **Awards:** 2026 ECSE Engineering Design Competition Winner, Dean's List 2025 (Top 5%)
- **Clubs/Societies:** Software Engineering Students Association (SESA), Web Development & Consulting Club (WDCC), Developers Society (DEVS)

Skills

Languages: TypeScript, Java, Python, C, C#
Frameworks & Libraries: Node.js, React, Express, Fastify, .NET, SpringBoot
Tools & Infrastructure: Git, PostgreSQL, SQLite, Supabase, MongoDB, Firebase, Cloudflare R2, AWS, Vercel

Experience

English and Mathematics Tutor, Thinking2Morrow – Auckland, NZ Feb 2024 – Present

- Tutored students from Year 1 to 13 across rigorous curricula, including NCEA, IGCSE, Cambridge, and IB
- Delivered personalized lesson plans in both one-on-one and group settings (managing up to 12 students per session)
- Developed strong professional communication, time management, and collaborative planning skills to adapt teaching methods to diverse learning preferences

Personal Projects

Tank Arena: Laser Tank Duels

- Placed first in the 2026 Engineering Design Competition, building an RC tank duelling game with infrared lasers.
- Built a local game server with Node.js and Fastify, achieving sub 10ms latency and 0% packet loss.
- Integrated local and cloud SQL database to serve a global leaderboard hosted on Vercel.
- Built a response mobile UI to emulate a GameBoy-inspired web-based controller.

Astral Images: Full-Stack Cloud Image Hosting Service

- Built a robust, full-stack cloud image hosting web application leveraging React, Express, and Cloudflare R2 for efficient blob storage
- Integrated Firebase for secure user authentication and session management
- Implemented automated CI/CD pipelines to streamline deployment workflows using Vercel

Ferrous Fellow: AI-Powered Interactive Toy Robot

- Designed and programmed an interactive toy robot that processes visual and verbal cues to recommend music for children
- Deployed machine learning models onto edge hardware by integrating TensorFlow Lite and OpenCV on a Raspberry Pi Pico
- Integrated the OpenAI API to dynamically analyze collected speech and facial data for personalized music recommendations

MeowPad: Custom Mechanical Number Pad

- Engineered a custom mechanical number pad from initial concept to manufactured hardware
- Designed the PCB schematics and layout using KiCad, and modeled the physical case using AutoCAD Inventor
- Developed custom C firmware using QMK for an Arduino Pro Micro, integrating an OLED display and rotary encoder